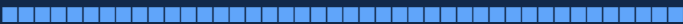


HUMORIX AI

Complete Developer Blueprint · v2.0



Architecture · UI Design · Conflict Handling · CBT/DBT Engine · Satvic Wellness

Document	Humorix AI — Complete Developer Blueprint (v2.0)
Contents	System architecture · Safety middleware · Fusion engine · DVITE protocol · UI design system · Conflict-handling logic · Satvic wellness delivery · Dynamic CBT/DBT intervention · Crisis resources · Ethics & roadmap
Tech Stack	Python · FastAPI · PyTorch · OpenCV · Librosa · Transformers · Streamlit · ONNX
Disclaimer	Supportive tool only. Not a substitute for licensed professional mental health care.

■ All content in this blueprint is derived from the Humorix AI Master Intelligence Document and associated technical specifications. Original authorship and intellectual property remain with the Humorix AI project.

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SECTION A

System Architecture

Five-layer pipeline · File structure · Real-time WebSocket flow

A1 Five-Layer Pipeline

Modular — any layer can be upgraded independently

#	Layer	Components	Function
1	INPUT	Webcam · Microphone · Text chat	Raw multimodal capture via WebSocket streams
2	PROCESSING	OpenCV · Librosa · HuggingFace Tokenizers	Frame extraction, audio features, text tokenisation
3	EMOTION	DeepFace/FER · Speech ML · Sentiment classifier	Per-modality emotion probability 0.0–1.0
4	DECISION	Fusion Engine · InterventionEngine · Satvic KB	Weighted fusion → severity → intervention selection
5	OUTPUT	FastAPI · Streamlit UI · Crisis escalation	Response delivery; helplines when threshold breached

A2 Project File Structure

```
humorix/
├── app/
│   ├── main.py # FastAPI entry point · WebSocket handlers
│   └── core/
│       ├── safety.py # Gatekeeper – first to touch every input
│       ├── fusion.py # Weighted Face/Voice/Text combination
│       ├── triage.py # DVITE policy engine – crisis vs. support
│       └── wellness.py # Satvic module (parallel, non-clinical)
├── models/
│   ├── vision/ # MediaPipe + DeepFace wrappers
│   ├── audio/ # Librosa feature extraction + speech model
│   └── nlp/ # Transformer sentiment/distress classifier
├── data/
│   ├── schemas.py # Pydantic: AcuityScore, UserSession, LogEntry
│   ├── protocols.py # All 8 disorder modules
│   └── resources.py # Crisis helpline contact objects
```

A3 Real-Time WebSocket Pipeline

Step	Component	Input	Output
1	safety.py	Raw text string	SAFE / CRITICAL flag

2a	vision/	Video frame (JPEG bytes)	Emotion dict {happy:0.7 ...}
2b	audio/	Audio chunk (PCM float32)	Emotion dict + arousal score
2c	nlp/	Tokenised text	Sentiment + distress category
3	fusion.py	Three dicts + base weights	fused_emotion · severity · disorder_risk
4	triage.py	Fusion output	Intervention text + escalation flag
5	main.py	Intervention response	JSON to UI / Streamlit

SECTION B

Safety & Processing Modules

Gatekeeper middleware · FER · Voice · NLP

B1 Safety Middleware — The Gatekeeper

Runs synchronously on EVERY input before any model inference

! This is the single most critical component. It intercepts all text before the LLM, emotion models, or any other system sees it. Zero exceptions. Target latency: under 5ms.

```
# app/core/safety.py

CRISIS_KEYWORDS = {"kill", "die", "suicide", "end it", "hurt myself", "can't go on"}

def run_safety_check(text_input: str) -> dict:
    """Synchronous gatekeeper — must complete in < 5ms."""
    if any(kw in text_input.lower() for kw in CRISIS_KEYWORDS):
        return {
            "status": "CRITICAL",
            "action": "TRIGGER_EMERGENCY_ESCALATION",
            "resources": "Tele-MANAS 14416 | KIRAN 1800-599-0019"
        }
    return {"status": "SAFE", "action": "PROCEED"}
```

B2 Facial Emotion Recognition (FER)

Libraries	OpenCV (capture) · MediaPipe (landmarks) · FER library · DeepFace
AUs Detected	Eyebrow (corrugator, frontalis) · Eye openness (levator) · Mouth curvature (zygomaticus) · Nose wrinkle · Head pose
Emotions	Happy · Sad · Angry · Fearful · Disgusted · Surprised · Neutral (Ekman 7)
Datasets	FER2013 (35,887 images) · AffectNet (~1M) · CK+ sequences
Confidence	0.0–1.0 per label. Score < 0.4 → auto-redistribute weight to voice + text
Performance	30 FPS (GPU) · 5–10 FPS (CPU fallback)
Bias Note	Audited per release across skin tone, age, gender, culture

B3 Voice Emotion Detection

Library	Librosa (Python audio analysis)
Features	MFCCs — voice timbre fingerprint Spectral Contrast — frequency range dynamics Pitch / Tone — fundamental frequency contour Speech Rate — words/min, pause patterns Tremor / Jitter — physiological stress marker
Outputs	Stress · Anxiety · Calmness · Sadness · Anger · Elevated mood

Panic Signal	Rapid speech + elevated pitch + irregular rhythm = panic pathway trigger
Datasets	IEMOCAP · DAIC-WOZ (clinical depression interviews with PHQ-8)

B4 Text / NLP Sentiment Analysis

Model	Transformer-based — BERT / RoBERTa family via HuggingFace
Categories	Positive · Negative · Neutral · Distress · Hopeless · Anger · Intrusive · Safety concern
Distress	Negative self-talk · Hopelessness · Social isolation · OCD loops · Worthlessness
Intent	Differentiates: venting ↔ seeking advice ↔ crisis disclosure
Override	Crisis keyword detected → skip all analysis, escalate immediately

SECTION C

Fusion & Decision Layer

Weighted combination · Override rules · Severity classification

C1 Weighted Fusion Engine

Base Weights	Face 0.45 · Voice 0.30 · Text 0.25
Fusion Method	Weighted average + rule-based overrides (see code below)
Override Rules	① Crisis keyword in text → bypass all weights, escalate immediately ② Any modality > 0.9 confidence → use max score (trust the signal) ③ Face confidence < 0.4 → redistribute face weight to voice + text
Output Schema	fused_emotion · severity · disorder_risk {} · overall_confidence

```
# app/core/fusion.py

def calculate_fusion(face: float, voice: float, text: float) -> float:
    # Weighted average
    fused = (face * 0.45) + (voice * 0.30) + (text * 0.25)

    # Override: one modality is very confident – trust it
    if max(face, voice, text) > 0.9:
        return max(face, voice, text)

    return fused
```

C2 Severity Classification & Routing

Severity	Criteria	System Response
none	Baseline / positive state	Psychoeducation, wellness tips, check-in
mild	Subclinical distress signals	Mild CBT technique, validation, grounding
moderate	Significant impairment markers	Moderate CBT + tracking + soft referral
severe	High distress + risk indicators	Severe intervention + hard referral + resource list
crisis	Immediate safety concern	CRISIS PROTOCOL — helplines first, no delay

SECTION D

DVITE Clinical Intervention Engine

CBT/DBT protocols · 8 disorder modules · Session tracking · Crisis resources

D1 The DVITE Protocol

01 DETECT	02 VALIDATE	03 INTERVENE	04 TRACK	05 ESCALATE
Fusion engine outputs emotion + severity score	Clinically-worded empathy response	CBT/DBT technique matched to severity tier	User rates 1-10 Trend computed	Threshold breach → helplines foregrounded

DETECT	Fusion engine returns fused_emotion + severity + disorder_risk probabilities.
VALIDATE	Clinically-worded statement normalising the experience. Never toxic positivity. Example: 'I hear that. What you're feeling makes sense given everything you're carrying.'
INTERVENE	Evidence-based technique from disorder-specific library, matched to severity tier. Randomised within tier to prevent repetition fatigue.
TRACK	User rates current intensity 1–10. Score appended to session_track[disorder]. After 3+ readings, trend direction computed: improving / stable / worsening.
ESCALATE	If severity is moderate/severe AND disorder_risk ≥ escalate_threshold: crisis resources provided unconditionally alongside intervention.

D2 Disorder Module Library — 8 Conditions

Code	Condition	Escalate At	Key Techniques by Severity
GAD	Generalised Anxiety	≥ 7/10	Mild: 5-4-3-2-1 grounding · Extended exhale Moderate: 4-7-8 breathing · PMR · Thought challenging Severe: CBT referral + KIRAN
PANIC	Panic Disorder	≥ 8/10	Mild: Box breathing · Floor grounding Moderate: Extended exhale · DARE response Severe: Crisis stabilisation + referral
MDD	Major Depressive Disorder	≥ 7/10	Mild: Behavioural activation · Gratitude · Opposite action Moderate: Pleasure prediction · Thought record · Social micro-action Severe: Safety inquiry + iCall / KIRAN
PTSD	Post-Traumatic Stress	≥ 7/10	Mild: Grounding anchor · Safe place visualisation Moderate: Window of tolerance · Titrated exposure Severe: EMDR + trauma-focused CBT referral
MANIA	Bipolar — Manic Episode	≥ 7/10	Mild: Sleep regulation reminder Moderate: 48-hour decision rule · Energy channelling Severe: Psychiatrist / care team contact
OCD	Obsessive-Compulsive	≥ 8/10	Mild: Thought defusion · 5-min compulsion delay Moderate: ERP ladder Severe: OCD specialist referral
ED	Eating Disorders	≥ 6/10	Mild: Body neutrality · Mindful eating Moderate: Critical voice challenging Severe: NAED helpline + specialist
ADHD	Attention Deficit Hyperactivity	≥ 8/10	Mild: 2-minute rule · Body doubling · Pomodoro Moderate: Task initiation hacks · Externalise everything Severe: Psychiatrist evaluation

D3 Session Tracking & Trend Analysis

Trend	Criterion	Action
Improving	Recent avg < Earlier avg by > 1 point	Affirm progress · maintain intervention
Stable	Recent avg within ± 1 of earlier	Continue approach · deepen check-in
Worsening	Recent avg > Earlier avg by > 1 point	Escalate severity tier · strengthen referral

✓ Minimum 3 user ratings collected before trend direction is computed.

D4 Crisis Protocol & Helplines

! When severity == 'crisis': validate feeling → one breathing instruction → foreground ALL resources below. No diagnosis. No delay. Unconditional.

India

- iCall (TISS) · 9152987821 — Free, trained counsellors
- Vandrevala Foundation · 1860-2662-345 — 24/7
- KIRAN Helpline · 1800-599-0019 — Govt, free, 24/7
- Tele-MANAS · 14416
- Snehi · 044-24640050 — Suicide prevention

International

- Crisis Text Line · Text HOME to 741741
- 988 Lifeline (US) · Call or text 988
- Samaritans (UK) · 116 123
- findahelpline.com — global directory

SECTION E

UI Design System

Adaptive visual language · Layout zones · Screen states · Design checklist

E1 Visual Language & Adaptive Design

The UI is a living system — it responds to the user's detected emotional state

Humorix's interface is not static. The Real-Time Adaptive UI requirement means the visual environment shifts based on the fusion engine output — using the brand's green-blue gradient as a base and adjusting saturation, pace, and density according to detected emotional state.

Emotional State	UI Behaviour	Rationale
Calm / Neutral	Standard vibrant green-blue gradient · Normal text speed · Full density	Baseline — no intervention required
Distress / Anxiety	Desaturated muted blue-grey · Text appears slowly · Reduced clutter	Slow, calm environment reduces overwhelm during grounding
ADHD / Mania (high energy)	Slightly more vibrant · Subtle motion elements · Short chunks	Match energy without overstimulating — direct, action-oriented
Crisis	Soft, warm, minimal · Crisis resources immediately visible · Large text	Remove all friction to accessing help

Typography	Clean sans-serif matching logo — clinical yet approachable. Single typeface throughout.
Colour System	Dynamic saturation/opacity based on detected emotion. Green-blue brand base.
Motion	Slow rhythmic animations during grounding. Subtle pulse for the emotion confidence indicator.
Accessibility	High contrast in crisis state. Slower text for anxiety. Larger tap targets on mobile.

E2 Layout — Three Zones

Zone	Component	Purpose
HEADER (top)	Status bar + disclaimer badge + cam/mic indicator	Always shows: 'Humorix AI is a supportive tool only.' Connection status always visible.
MAIN VIEW (centre)	Adaptive Canvas — AI response + organic waveform visualisation	The Fusion Engine confidence is shown as a soft pulsing circle. Text speed adapts to state.
SIDEBAR (left)	Holistic Hub: Meal Plans · Breathing library · Session history	Quick access to Satvic plans, CBT/DBT exercises, and past mood tracking.
FOOTER (bottom)	Text input · Voice toggle · Emergency (crisis) button	Crisis button is always visible, always reachable — persistent floating action button.

E3 Key Functional Components

The Persistent Crisis Button

A floating action button (FAB) or footer button — always present, always reachable. One tap triggers the CRISIS PROTOCOL: breathing stabilisation + all helplines foregrounded immediately. No confirmation dialogs, no delays.

The Pulse — Multimodal Confidence Indicator

A soft, circular visual element in the centre of the screen that pulses based on overall_confidence from the fusion engine. Low confidence: neutral grey pulse. High confidence: reflects the detected emotion's colour. Communicates that the AI is 'listening' without being invasive or surveillance-feeling.

Satvic Sidebar Cards

Recipes and wellness interventions appear as clean cards with ingredient checklists. When Tamasik (lethargy/depression) is detected, the sidebar automatically surfaces a relevant Satvic meal plan or grounding activity — but as a gentle notification badge, not an interruption.

Privacy Controls

Multimodal Toggle	Switch between: voice-only · text-only · full multimodal (face/voice/text)
Delete Data	Clear 'Stop Processing' toggle — immediately halts all biometric analysis
Consent Modal	Camera and microphone cannot activate until user has acknowledged consent

E4 Four Screen States

State	Visual	Content
State 1 Check-in (Standard)	Bright, welcoming. Standard gradient.	Prompt: 'How are you feeling today?' · Full chat interface · Sidebar available
State 2 Grounding (Anxiety/Panic)	Muted blue-grey. Slow expanding/contracting circle guides breathing. Text appears gradually.	Breathing exercise foreground · Text paced to reduce overwhelm · Crisis button prominent
State 3 Wellness (Satvic focus)	Calm, nature palette. Soft greens and earth tones.	Satvic meal plan relevant to mood · Recipe cards · Mindful eating prompts
State 4 Crisis	Minimal, warm, large text. All non-essential UI removed.	Crisis helplines immediately visible · One breathing instruction · No therapy content until stable

E5 Design Checklist

Item	Requirement	Status
Disclaimer	Visible on every screen, every session	Required
Crisis FAB	Persistent, one-tap access — never hidden	Required
Consent Modal	Camera/mic blocked until acknowledged	Required
Multimodal Toggle	Voice-only / text-only / full multimodal modes	Required
Privacy Control	'Delete Data' / 'Stop Processing' always accessible	Required

Adaptive Colour	UI palette shifts with detected emotional state	Required
Pulse Indicator	Soft confidence visualiser — not surveillance-feeling	Required
Text Speed Control	Slower text delivery for anxiety states	Required
Satvic Cards	Clean cards with ingredient checklists	Recommended
Session History	Mood trend chart accessible in sidebar	Recommended

SECTION F

Conflict Handling Logic

When the user says OK but biometrics say distress

F1 The Core Conflict Scenario

A fundamental challenge in multimodal emotion AI is the data conflict: the user explicitly reports they are fine ('I'm okay'), but facial and voice biometrics indicate significant distress. How the system responds to this gap is critical — both for clinical safety and user trust.

Option	Approach	Risk	Score
A — Empathetic Inquiry	Gently acknowledge the discrepancy: 'I hear you're okay, but I'm noticing some tension — I'm here if you want to talk.'	User may still deny (but trust is preserved)	Recommended ✓
B — Clinical Neutrality	Focus on physiological data: 'I'm detecting indicators of distress. Would you like a grounding exercise?'	Feels robotic; may damage rapport	Secondary
C — System Override	Treat biometrics as truth; skip conversation, launch protocol immediately.	Violates user agency; high dropout risk; ethically problematic	Avoid

F2 Empathetic Inquiry Protocol

Option A — aligns with Multimodal Fusion and the DVITE Validate step

◆ Clinical research shows that respecting user self-report while gently surfacing biometric discrepancies reduces denial-driven dropout and improves disclosure rates. The fusion engine logs the conflict for professional review without overriding the user.

Conflict Level	User State	Bio State	Response Strategy	Example
Low	Text: OK	Bio: Mild	Light validation	'Sounds good — tension easing?'
Medium	Text: OK	Bio: High	Empathetic Inquiry	'I hear you're okay, but I'm detecting tension in your expression — I'm here.'
High	Text: OK	Bio: Crisis	Inquiry + Helplines	Empathetic inquiry + pulsing crisis button + soft helpline mention

F3 Code Implementation

```
# app/core/fusion.py – conflict handler

def handle_conflict(user_text: str, bio_score: float) -> dict:
    sentiment = analyze_sentiment(user_text) # Transformers pipeline
    # Conflict: bio signals distress, text says OK
    conflict = bio_score > 0.7 and sentiment < 0.5

    if conflict:
        return {
```

```
"mode": "empathetic_inquiry",  
"response": "I hear you're okay – I'm also noticing some tension.  
I'm here if you want to talk about what's going on.",  
"log": "conflict_medium", # Logged for pro review dashboard  
"next": "offer_grounding" # Passive, not forced  
}  
return {"mode": "standard", "response": "Great to hear."}  
  
# The disclaimer is automatically prepended on all conflict responses
```

SECTION G

Satvic Wellness Delivery

Activation rules · Wellness Nudge method · Meal plans · 9 Food Laws

G1 Module Activation Rules

■ When DVITE severity_score > 0.6 (Moderate–Severe), Satvic recommendations are automatically suppressed. This prevents 'wellness washing' — never suggest a juice recipe during a panic attack.

File	app/core/wellness.py — independent, never touches DVITE logic
Enabled	severity_score ≤ 0.6 → wellness content active
Disabled	severity_score > 0.6 → wellness content suppressed, clinical takes priority
Scope	Meal recommendations · Food law reminders · Mindful eating cues · Satvic recipes

G2 Delivery Method — The Wellness Nudge

Contextual / soft delivery — woven into conversation, not prescribed

Rather than presenting Satvic interventions as a clinical prescription or a sidebar notification badge, the recommended approach is the Wellness Nudge: weaving the recommendation conversationally, framed around the user's current detected state. This validates first, then suggests — maintaining the 'supportive tool' positioning.

Method	Description	Adherence	Use
Prescription Card	Direct: 'Based on Rajasic signals, I recommend Ash Gourd juice.'	Moderate	Avoid — feels like doctor's orders
Wellness Nudge ✓	Contextual: 'Your energy feels scattered. A grounding kitchari might help center you?'	High	Recommended
Sidebar Badge	Non-intrusive badge: 'A Satvic adjustment is available.'	Low	Secondary — low visibility

```
# app/core/wellness.py

NUDGES = {
    'rajasic': 'Your energy feels scattered. A grounding kitchari (mung dal + rice + cumin) might help bring you back to centre?',
    'tamasic': 'Detected low vitality. Fresh ash gourd juice could refresh — blend with ginger, sip slowly.',
    'balanced': 'You seem centred today. Keep it up.'
}

def satvic_nudge(guna_state: str) -> str:
    return NUDGES.get(guna_state, NUDGES['balanced'])
```

G3 Three Meal Plans — Overview

Plan	For Whom	Key Rule	Duration
HEALING	Chronic condition reversal (thyroid, diabetes, PCOD, acne, fatty liver)	No grains at dinner · Fruit-only breakfast · 14–16 hr fasting window	Max 3 months
LIFESTYLE	Sedentary lifestyle / weight-gain tendency	Grain + legume at lunch · Optional grain at dinner	Ongoing
ACTIVE	Underweight or intense daily exercise	Grain + legume at breakfast AND lunch · Higher caloric density	Ongoing

Healing Plan — Daily Schedule

Time	Meal	Options
08:00	Detox Juice	Ash Gourd (400ml) · Coconut Water · Glowing Green Juice · ABC Juice · Bathua Tomato (winter)
10:00	Fruit Breakfast	Single seasonal fruit · Mixed fruits · Smoothie Bowl · Pure Satvic Salad · Clear Soup (winter) Diabetics: guava / papaya / pear / apple only — no smoothies
13:00	Grain Lunch	Satvic Roti · Millet Roti · Satvic Khichadi (1 cup rice + 5 cups veg) · Daliya · Millet Upma Rule: 1 grain/legume only + 2–3× more vegetables
16:00	Optional Mid-Meal	Fresh juice · Herbal tea · 6–8 soaked nuts · Fresh coconut — skip if not genuinely hungry
19:00	Veggie Dinner (NO grains)	Salads: Carrot Raisin · Thai Papaya · Zesty Beet · Cheesy (cashew) Soups: Pumpkin · Spinach · Pea Carrot · Broccoli Potato · Tomato · Carrot Cumin

✓ Fasting window: 14–16 hours between dinner (19:00) and morning juice (08:00). Essential for autophagy and hormone regulation.

G4 9 Satvic Food Laws — Quick Reference

#	Law	Avoid	Eat Instead
1	No Dead Foods	Packaged / tinned / canned / bottled	Fresh fruits, vegetables, grains — farm to kitchen
2	No Refined Foods	White sugar, flour, rice, refined oil	Dates, jaggery, millets, brown rice, cold-pressed oils
3	No Animal Products	Meat, fish, eggs, dairy, ghee, paneer	Coconut milk, almond milk, cashew cheese (all homemade)
4	Less Grain, More Veg	Grain-dominant plates	2× more vegetables than grain/legume on every plate
5	Seasonal & Local	Imported exotics (blueberries, kale, macadamia)	Regional seasonal produce — cheaper, more bioavailable
6	Soak Nuts	Unsoaked nuts — enzyme inhibitors	Soak 6–8 hours, discard water, then use
7	Mild Spices Only	Packaged garam masala, red chili powder	Fresh green chili, black pepper, curry leaves, herbs
8	Minimal Cooking	High-heat frying, prolonged boiling	Lowest temperature, shortest time — raw where possible
9	No Stimulants	Tea, coffee, onions, garlic	Herbal teas (tulsi, lemongrass, rose), coconut water

SECTION H

Dynamic Intervention Engine

CBT vs DBT selection · Severity-based routing · Code blueprint

H1 CBT vs. DBT — Selection Logic

The Dynamic Hybrid automatically selects the right approach for the severity level

A single intervention method cannot serve all severity levels effectively. The Dynamic Hybrid approach uses the fusion engine's severity score to route: Cognitive Behavioural Therapy (CBT) builds long-term resilience at low distress levels, while Dialectical Behaviour Therapy grounding (DBT) stabilises the nervous system immediately at high distress levels — bypassing analytical cognition when it is least available.

Distress Level	Score Range	Intervention	Mechanism	Goal
Low	0.0 – 0.4	CBT — Thought Record	Analytical: label distortion → evidence challenge → balanced alternative	Build long-term cognitive resilience
Medium	0.4 – 0.7	Hybrid Introduction	Brief grounding first (stabilise) → then light CBT intro	Stabilise + begin building skills
High	0.7 – 1.0	DBT — Grounding Sequence	Sensory: 5 things seen, 4 touched, 3 heard, 2 smelled, 1 tasted	Immediate nervous system reset
Crisis	> crisis threshold	Stabilisation + Escalation	One breathing instruction → helplines foregrounded immediately	Safety first, no therapy until stable

H2 Severity-Based Routing — Full Detail

Severity	Trigger Example	Response	Next Step
Low (0–0.4)	Mild rumination · 'I keep thinking about it'	Thought Record: 'That sounds like catastrophising. What's the evidence for and against?'	Track intensity · offer journal
Medium (0.4–0.7)	Mixed signals · some tension	Hybrid: brief grounding → 'Now, what's one thought driving this feeling?'	Validate insight · soft referral mention
High (0.7+)	Panic biometrics · rapid speech + face tension	5-4-3-2-1 grounding: '5 things you can see right now?'	Re-scan biometrics post-exercise · escalate if no improvement
Crisis	Crisis keyword OR biometrics extreme	Validate + breathe + helplines immediately	No therapy content until user is stabilised

H3 Code Blueprint — Dynamic Intervention Selector

```
# app/core/triage.py

def select_intervention(severity: float, emotions: dict) -> dict:

    # HIGH distress – DBT grounding (bypass analytical thought)
```

```
if severity > 0.7:
    return {
        "type": "dbt_grounding",
        "script": "5 things you see? 4 you can touch? 3 you hear? 2 smell? 1 taste.",
        "next": "rescan_biometrics" # Check if grounding worked
    }

# MEDIUM distress – hybrid
elif severity > 0.4:
    return {
        "type": "hybrid",
        "script": "Let's take one slow breath first... Now, what's one thought
that's driving this feeling?",
        "next": "validate_and_deepen"
    }

# LOW distress – CBT thought record (build resilience)
else:
    thought = emotions.get('dominant_thought', 'this situation')
    return {
        "type": "cbt_thought_record",
        "script": f"That sounds like it might be {thought}.
What's the evidence for, and against, that thought?",
        "next": "record_and_track"
    }

# Usage in pipeline
result = select_intervention(severity=0.82, emotions={'dominant_thought': 'catastrophising'})
```

SECTION I

Deployment & Ethical Framework

Deployment modes · Privacy design · Non-negotiables · Limitations · Roadmap

I1 Deployment Modes

Mode	Stack	When to Use
Local Dev	Python + Streamlit · CPU/GPU	Development, personal use — full control, fully private
ONNX (Edge)	ONNX Runtime · CPU inference	Production start — lower cost, reduced latency, no GPU required
GPU Cloud	Docker + NVIDIA Triton · AWS/GCP/Azure	Scale when load exceeds local capacity
Mobile	ONNX → React Native / Flutter	Consumer mobile — models compressed and exported
Clinical API	FastAPI REST · JWT auth	EHR/EMR integration or telehealth platform partnership

I2 Privacy Design — Ephemeral Processing

! Biometric data (video frames, audio chunks) MUST be discarded immediately after the fusion score is computed. Do not store raw streams without explicit consent.

Ephemeral Rule	Face + audio discarded after fusion score extracted. Never persisted.
Text Logs	Session text only with explicit opt-in consent. Encrypted at rest.
Consent Modal	Camera/mic blocked until user has acknowledged consent modal.
Data Deletion	Users can delete all session data at any time. Completes within 24h.
Third-Party	No biometric data shared with or sold to any third party.

I3 Ethical Non-Negotiables

Principle	Implementation
No Diagnosis	Never outputs a clinical diagnosis. Identifies risk patterns only.
Always Disclaim	Disclaimer shown every session. Cannot be disabled by any user or developer.
Crisis First	When crisis detected, all other content is secondary. Helplines appear immediately.
Informed Consent	Explicit opt-in before any camera or microphone data is captured.
Transparency	System explains what it detected and why it responded — no black-box outputs.
Bias Audit	Demographic performance audits mandatory before each model release.
No Data Sale	User data never sold, shared, or used for external commercial model training.

I4 Known Limitations & Future Roadmap

Known Limitations

Limitation	Detail	Current Mitigation
Demographic FER gaps	Models skewed toward lighter skin / Western subjects	Dataset expansion + audit per release
Environment dependency	Degrades in poor light, unusual angles, occlusion	Fallback to voice + text when face confidence < 0.4
Discrete categorisation	Real emotions are continuous; models output discrete labels	Confidence scores surfaced; multi-label output planned
No cross-session memory	Each session starts fresh	Federated memory system in roadmap
English-primary NLP	Transformers trained mainly on English	Hindi, Tamil, Telugu, Bengali, Marathi models planned
No clinical validation	No large-scale RCT yet	NIMHANS partnership in roadmap

Future Roadmap

PERSONALISED AI	WEARABLES	EMOTIONAL MEMORY	ADAPTIVE UI	CLINICAL TRIALS	MULTILINGUAL
Initiative	Description				Priority
Personalised AI	Federated learning per user — no raw biometric transmission				High
Wearable Integration	Heart rate, GSR, sleep trackers enrich physiological layer				High
Emotional Memory	Cross-session mood patterns, trigger tracking, progress				High
Real-Time Adaptive UI	Interface adjusts pace, colour, density per emotional state				Medium
Clinical Validation	NIMHANS-partnered RCT vs. baseline care				High
Indian Language NLP	Hindi, Tamil, Telugu, Bengali, Marathi full support				High
Satvic Personalisation	Auto meal plan from emotion state + health + season				Medium

DETECT → VALIDATE → INTERVENE → TRACK → ESCALATE

Humorix AI · Emotionally intelligent · Ethically grounded · Holistically designed